

Method of Moments Estimator

The k -th (uncentered) moment of a random variable X is defined by

$$\mu'_k = E(X^k)$$

which is in general a function of the parameters of the distribution function of X . The k -th *sample moment* of a random sample of size T is defined by

$$m'_k = \frac{1}{T} \sum_{i=1}^T X_i^k$$

where $m'_1 = \bar{X}$, the sample mean.

The basic approach of the method of moments is to set the sample moments of lowest order with the moments of the population distribution, and then solves for the unknown parameters. For example, if there is only one unknown parameter, it is derived from the first moment condition $\mu'_1 = m'_1$. If there are two parameters, they are derived from the first and second moment conditions: $\mu'_1 = m'_1$ and $\mu'_2 = m'_2$. These estimators are called the *method of moment estimator* (MME).

Example. A Bernoulli density $f(x; \theta) = \theta^x(1-\theta)^{1-x}$ has one parameter θ , and has the mean $E(X) = \theta$. Hence, the MME of θ can be derived from the first moment condition $\mu'_1 = m'_1$, which gives $\hat{\theta} = \bar{X}$. $\blacksquare$

Example. A normal distribution has two parameters, mean μ and variance σ^2 . Since $E(X^2) = \sigma^2 + \mu^2$, the MME of μ and σ^2 are derived from the first two moments

$$\mu'_1 = m'_1 \Rightarrow \hat{\mu} = \bar{X}$$

$$\mu'_2 = m'_2 \Rightarrow \hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{T} \sum_{i=1}^T X_i^2 \Rightarrow \hat{\sigma}^2 = \frac{1}{T} \sum_{i=1}^T X_i^2 - \hat{\mu}^2 = \frac{1}{T} \sum_{i=1}^T (X_i - \bar{X})^2 \quad \blacksquare$$

One should notice that the normality of the distribution in this example is irrelevant to the estimators of the mean and variance. In other words, these estimators of mean μ and variance σ^2 are their MME's for any population distribution whose moments exist and the two parameters are independent parameters. This implies that, unlike the MLE, the MME does not require the knowledge of the distribution function as long as the moment conditions are known.

We just described the method of moments as equating the population moments to the corresponding sample moments of proper order. The method can also be described as representing the *population mean* of a function of random variables by the *sample mean* of the same function, where the function involves unknown parameters. As a trivial example, consider the functions $(X - \mu)$ and $[(X - \mu)^2 - \sigma^2]$. By the definition of the mean and variance, population means of these functions are zeros:

$$E(X - \mu) = 0, \quad E[(X - \mu)^2 - \sigma^2] = 0$$

These equations are called the *population* (or *theoretical*) *moment conditions*. The method of moment replaces these population means by their sample means

$$\frac{1}{T} \sum_{t=1}^T (X_t - \hat{\mu}) = 0, \quad \frac{1}{T} \sum_{t=1}^T [(X_t - \hat{\mu})^2 - \hat{\sigma}^2] = 0$$

which are called the *sample* (or *empirical*) *moment conditions*. The MME of μ and σ^2 are the solutions to the sample moment conditions. The estimators are of course same as those derived earlier.

The population moment condition can arise from the specification of the distribution function or other properties of the model under consideration. We consider three simple examples below.

MME of a Standard Linear Regression Model

The assumptions in a standard linear regression model $y_t = X_t\beta + u_t$ are $E(u_t|X_t) = 0$ and $E(u_t^2|X_t) = \sigma^2$. The first condition of u_t being uncorrelated with regressors implies the unconditional moments $E(X_t' u_t) = 0$ ¹. Therefore, we have population moment conditions

$$E(X_t' u_t) = 0, \quad E(u_t^2) = E[E(u_t^2|X_t)] = \sigma^2$$

The second condition restricts the error terms to be homoscedastic. The corresponding sample moment conditions are

$$\begin{aligned} E(X_t' u_t) = 0 &\Rightarrow \frac{1}{T} \sum_{t=1}^T X_t' u_t = \frac{1}{T} \sum_{t=1}^T X_t' (y_t - X_t\beta) = \frac{1}{T} (X'y - X'X\beta) = 0 \\ E(u_t^2|X_t) - \sigma^2 = 0 &\Rightarrow \frac{1}{T} \sum_{t=1}^T (u_t^2 - \sigma^2) = \frac{1}{T} \sum_{t=1}^T (y_t - X_t\beta)^2 - \sigma^2 = \frac{1}{T} (y - X\beta)'(y - X\beta) - \sigma^2 = 0 \end{aligned}$$

We have $K+1$ unknown parameters β and σ^2 , and these sample moment conditions provide exactly $K+1$ independent equations. The solution for β gives an estimator that is exactly same as the OLSE. Thus, the OLSE of β is its MME based on the moment condition $E(X_t' u_t) = 0$, which is imposed by the model specification². The MME of σ^2 is same as the MLE $\hat{\sigma}^2 = \hat{u}'\hat{u}/T$.

Moving Average Model

Consider a moving average MA(1) model

$$y_t = \epsilon_t + \alpha\epsilon_{t-1}, \quad \epsilon_t \sim i.i.d.(0, \sigma^2)$$

which has theoretical moments

$$E(y_t) = 0, \quad E(y_t^2) = (1 + \alpha^2)\sigma^2, \quad E(y_t y_{t-1}) = \alpha\sigma^2$$

There are two parameters α and σ^2 . Since the first moment does not involve any parameter, we use the last two moment conditions and consider their sample moment conditions

$$\frac{1}{T} \sum_{t=1}^T (y_t^2 - (1 + \alpha^2)\sigma^2) = 0, \quad \frac{1}{T} \sum_{t=1}^T (y_t y_{t-1} - \alpha\sigma^2) = 0$$

¹ This is based on the fact that the unconditional mean is equal to the mean of a conditional mean: $E(X_t' u_t) = E[E(u_t|X_t)] = 0$.

² When the moment condition $E(X_t' u_t) = 0$ does not hold as in a simultaneous equations model, the moment condition between a set of instrumental variable W_t and error term, $E(W_t' u_t) = 0$, is used.

Let the estimate of autocorrelation function be

$$\hat{\rho} = \frac{\sum y_t y_{t-1}}{\sum y_t^2}$$

Then, the sample moment conditions can be written as

$$\hat{\rho} \alpha^2 - \alpha + \hat{\rho} = 0$$

The MM estimator of α is the solution of this quadratic equation, where $|\hat{\rho}| \leq 1/2$ is required for a real solution.

MLE as a MME

The maximum likelihood estimator can also be considered as a MME. Let $\mathbf{y}' = (y_1, y_2, \dots, y_T)$ be a random sample from a population characterized by a density $f(y_i; X_p, \theta)$. The log likelihood function for the sample is

$$\ln L(\mathbf{y}; X, \theta) = \sum_{i=1}^T \ln f(y_i; X_p, \theta)$$

The MLE of θ is found as the solution to the first order condition

$$\frac{\partial \ln L(\mathbf{y}; X, \theta)}{\partial \theta} = \sum_{i=1}^T \frac{\partial \ln f(y_i; X_p, \theta)}{\partial \theta} = \sum_{i=1}^T g_i(\theta) = 0$$

where $\mathbf{g}_i(\theta)$ is the gradient vector. We have shown earlier (Proposition 3 in MLE) that $E[\mathbf{g}_i(\theta)] = \mathbf{0}$. This is a population moment condition, and its corresponding sample moment condition is

$$\frac{1}{T} \sum_{i=1}^T \mathbf{g}_i(\theta) = \mathbf{0}$$

the solution of which is the MME of θ . This is the same as the first order condition for the MLE.

Homework #4

(a) You found the MLE of the parameters of double exponential distribution in question (3) of Homework #3. Suppose you don't know the distribution function, but you know only the first two moments of a random sample X_i of sample size n : $E(X_i) = \mu$ and $\text{var}(X_i) = 2\beta^2$. Find the MME of μ and β , and compare them with their MLE's.

Generalized Method of Moment (GMM) Estimator

These examples illustrate the MME in the *exactly (just)* identified case, in which the number of unknown parameters is the same as the number of sample moment equations, and there is a unique solution for the parameters. In some cases we may have more population moment conditions than the number of parameters, so that the system of sample moment conditions is *overidentified*, or *overdetermined*. For example, consider the example of the MME of the standard linear regression model. We used the weak exogeneity condition $E(u_t|X_t) = 0$ for all t . Suppose we wish to use the assumption of strong exogeneity of regressors, $E(u_t|X_s) = 0$ for all t and s . Then, we can use the theoretical moments conditions $E(u_t|X_t) = 0$ and $E(u_t|X_{t-1}) = 0$, which imply $E(X_t'u_t) = 0$ and $E(X_{t-1}'u_t) = 0$. This gives a system of $2K$ equations for K unknown parameters β , and there is no solution that satisfies all moment conditions. The generalized method of moment (GMM) is designed to estimate the parameters in such a case. Of course, the case of just-identified moment conditions is a special case of the GMM.

To introduce the ideas of the GMM, we will discuss the GMM estimator of a linear regression model

$$y_t = Z_t'\theta_0 + u_t, \quad t=1,2,\dots,T \quad \Rightarrow \quad y = Z\theta_0 + u$$

where θ_0 is the true parameter value, Z_t is a $1 \times p$ vector of regressors, some of which may be correlated with u_t . The covariance matrix of the error vector is denoted by $\text{cov}(u|X) = (\omega_{tt}) = \Omega$, where Ω can be a scalar matrix (homoscedasticity and no serial correlations), a diagonal matrix (heteroscedasticity and no serial correlations), or a general symmetric positive matrix (heteroscedasticity and serial correlations). We denote by X_t the set of K (weakly) exogenous variables, which include all exogenous variables in Z_t . Some notations are listed below for an easy reference:

$$m_t = X_t'u_t = X_t'(y_t - Z_t'\theta)$$

$$\mu_t(\theta_0) \equiv E(m_t) = E(X_t'u_t) = E[X_t'(y_t - Z_t'\theta_0)] = 0 \quad \dots\dots\dots \text{Population moment condition}$$

$$\bar{m}(\theta) = \frac{1}{T} \sum_{t=1}^T m_t = \frac{1}{T} \sum_{t=1}^T X_t'u_t = \frac{1}{T} X'u = \frac{1}{T} X'(y - Z\theta) \quad \dots\dots\dots \text{Sample moment condition}$$

$$\bar{\mu}(\theta_0) = T^{-1} \sum_{t=1}^T \mu_t(\theta_0)$$

$$\Sigma_{xz} = E[X_t'Z_t]$$

Identifiability of Parameters

Exogeneity of X_t gives the population moment condition $\mu_t(\theta_0) = 0$. This says that the moment condition is satisfied by the true value θ_0 of the parameter. Is there another value θ_1 that satisfies the moment condition?

Parameter θ_0 is said to be *identifiable by the population moment condition*¹ if and only if $\mu_t(\theta)=0$ has a unique solution for θ and the solution is equal to the true value θ_0 . If there is at least one other solution θ_1 that satisfies the population moment condition ($\mu_t(\theta_1)=0$), then parameter vector θ_0 is said to be *unidentifiable by the population moment condition*. If θ_0 is not identifiable and there are more than one θ that satisfies the population moment condition beside the true parameter, we would not know which parameter we estimate by using the sample moment that matches the population moment condition.

What conditions are required to assure identifiability of θ_0 by the population moment condition? By adding and subtracting, we can rewrite the moment condition as

$$\begin{aligned}\mu_t(\theta) &= E[X_t'(y_t - Z_t\theta)] = E[X_t'(y_t - Z_t\theta_0)] + E[X_t'(y_t - Z_t\theta)] - E[X_t'(y_t - Z_t\theta_0)] \\ &= 0 + E[X_t'Z_t](\theta - \theta_0) \equiv \Sigma_{xz}(\theta - \theta_0) = 0\end{aligned}$$

The last expression indicates that the solution for this moment condition occurs at $\theta = \theta_0$ if and only if Σ_{xz} has a full column rank: $\text{rank}(\Sigma_{xz}) = p$. If this rank condition is satisfied, the only solution to the equation above is $(\theta - \theta_0) = 0$, i.e., $\theta = \theta_0$. If $\text{rank}(\Sigma_{xz}) < p$, there is a non-zero vector c such that $\Sigma_{xz}c = 0$ so that $(\theta - \theta_0) = c$. This gives $\theta = \theta_0 + c \neq \theta_0$. Since Σ_{xz} has a dimension of $K \times p$, the *necessary condition* for the full column rank is $K \geq p$.

Lemma. The necessary condition (order condition) for the identifiability of θ_0 by the population moment condition is $K \geq p$. The necessary and sufficient condition (rank condition) for the identifiability of θ_0 by the population moment condition is $\text{rank}[E(X_t'Z_t)] = p$.

If the rank condition is satisfied, θ_0 is said to be *just-identified* if $K=p$, and *overidentified* if $K > p$.

GMM Estimation Procedure

If parameter θ_0 is identifiable, the GMM finds its estimate from the sample moment condition $\bar{m}(\theta) = 0$. This is a system of K equations in p unknown parameters. If $p=K$, it is a just-identified system and $X'Z$ is a nonsingular square matrix. Hence, we can compute θ from $\bar{m}(\theta) = 0$:

$$\bar{m}(\theta) = \frac{1}{T} X'(y - Z\theta) = 0 \Rightarrow X'y - X'Z\theta = 0 \Rightarrow \hat{\theta} = (X'Z)^{-1}X'y$$

This estimator is called the instrumental variable (IV) estimator and X is called the instrumental variables. This estimator gives a perfect fit of the sample moment condition and $\bar{m}(\hat{\theta})$ is identically zero for all elements.

In the case of overidentifying moment conditions ($K > p$), there is no solution that satisfies all moment conditions. That is, we cannot find an estimator such that $\bar{m}(\hat{\theta}) = 0$ for all elements. All that we can do is one

¹ The term "by the population moment condition" is emphasized here because, in a conventional simultaneous equations model, the identifiability of parameters is achieved by imposing zero or other restrictions on the coefficients.

of the following two approaches:

- (i) to reduce the number of moment condition to the number of just identified case, or
- (ii) to find an estimator that makes a certain measure of the distance of $\bar{m}(\hat{\theta})$ from zero as small as possible.

(i) In the first approach, we reduce the overidentified system to the just-identified system by selecting p number of sample moment conditions and then solve for θ . Let F be a $K \times p$ selection matrix with full column rank. If we wish to pick just the first p conditions, we specify $F' = (I_p, 0)$. This particular selection matrix is not a sensible choice as we discard some of the moment conditions and picking the first p conditions is arbitrary. A more sensible approach is to find the **best** p linearly independent linear combinations of K moment conditions, where the "best" selection matrix is the selection matrix that leads to the best estimator of θ . Premultiplying F' to the sample moment conditions, we have the just identified p moment conditions

$$\tilde{m}(\theta) = F' \bar{m}(\theta) = \frac{1}{T} (XF)' (y - Z\theta) = \frac{1}{T} (F' X' y - F' X' Z\theta) = 0$$

Since $F' X' Z$ is a square matrix of full rank, the solution to this set of conditions is

$$\tilde{\theta} = (F' X' Z)^{-1} F' X' y = \theta + (F' X' Z)^{-1} F' X' u$$

with an asymptotic covariance matrix

$$\text{cov}(\tilde{\theta} | X) = (F' X' Z)^{-1} F' X' \Omega X F (Z' X F)^{-1}$$

Now, how do we find the best selection matrix? Note that the variance and covariance matrix of $\tilde{\theta}$ depends on the choice of weighting matrix. This suggests that we wish to choose F that yields the highest precision of the estimator, that is, the smallest covariance matrix. It is straightforward to prove (see the homework question) that the optimal weighting matrix is

$$F = (X' \Omega X)^{-1} X' Z$$

This choice gives the estimator

$$\hat{\theta} = [Z' X (X' \Omega X)^{-1} X' Z]^{-1} Z' X (X' \Omega X)^{-1} X' y$$

with the asymptotic covariance matrix

$$\text{cov}(\hat{\theta} | X) = [Z' X (X' \Omega X)^{-1} X' Z]^{-1}$$

The difference in the covariance matrices, $\text{cov}(\tilde{\theta}) - \text{cov}(\hat{\theta})$, is positive semidefinite for any choice of F for $\tilde{\theta}$.

Remarks:

- (a) Premultiply X' to the regression model under study to get

$$X' y = X' Z \theta + \epsilon, \quad \epsilon = X' u, \quad \text{cov}(\epsilon) = X' \Omega X$$

The GMM estimator $\hat{\theta}$ is the GLS estimator of this transformed regression model.

- (b) If the error term has a scalar covariance matrix, $\text{cov}(u | X) = \sigma^2 I$, then the GMM estimator becomes

$$\hat{\theta}_{2SLS} = [Z' X (X' X)^{-1} X' Z]^{-1} Z' X (X' X)^{-1} X' y$$

This estimator is called the *two-stage least squares (2SLS) estimator* because it can be obtained in two stages:

Stage 1: Regress Z on X , and obtain the predicted value of Z : $\hat{Z} = X(X'X)^{-1}X'Z$

Stage 2: Regress y on $\hat{Z}$: $\hat{\theta}_{2SLS} = (\hat{Z}'\hat{Z})^{-1}\hat{Z}'y$

Homework #4

(b) Consider the covariance matrices of the estimators with weighting matrix $F = (X'\Omega X)^{-1}X'Z$ and any other weighting matrix F :

$$\text{cov}(\hat{\theta}) = [Z'X(X'\Omega X)^{-1}X'Z]^{-1}$$

$$\text{cov}(\tilde{\theta}) = (F'X'Z)^{-1}F'X'\Omega XF(Z'XF)^{-1}$$

Show that $\text{cov}(\tilde{\theta}) - \text{cov}(\hat{\theta})$ is positive semidefinite.

Hint: Show $[\text{cov}(\hat{\theta})]^{-1} - [\text{cov}(\tilde{\theta})]^{-1}$ is positive semidefinite. The Cholesky decomposition $\Omega = CC'$ may be useful.

(ii) The second approach to the problem of overidentification is to define the distance of $\bar{m}(\theta)$ from zero by the squared Euclidian norm with an adjustment for the differences in the variances in $\bar{m}(\theta)$. The general form of the **objective (criterion) function for the GMM** is the quadratic form in the sample moment $\bar{m}(\theta)$

$$\min_{\theta} J(\theta) = T[\bar{m}(\theta)]' W[\bar{m}(\theta)] = T[T^{-1}X'(y - Z\theta)]' W[T^{-1}X'(y - Z\theta)]$$

where W is a $K \times K$ positive definite symmetric matrix which converges to a nonrandom positive definite matrix. The solution to this minimization problem is the *Generalized Method of Moment (GMM)* estimator.

Remarks:

(a) We need to choose W and the best choice of W is the one that yields the *efficient* estimator of θ .

(b) Multiplication of T in the objective function does not have any effect on the minimization, but it is useful to define it this way because $J(\hat{\theta})$ is used as the test statistic for the test of overidentifying moment conditions, which is discussed later.

Since $\partial \bar{m}(\theta) / \partial \theta' = -X'Z/T$, the first order condition for the minimization of $J(\theta)$ is

$$\frac{\partial J}{\partial \theta} = -2T[T^{-1}Z'X]W[T^{-1}X'(y - Z\theta)] = -2T^{-1}Z'XWX'(y - Z\theta) = 0$$

which can be solved for θ

$$\hat{\theta} = (Z'XWX'Z)^{-1}Z'XWX'y$$

Note that this estimator is the same as the GMM estimator discussed in the first approach if we choose

$$W = (X' \Omega X / T)^{-1},$$

Asymptotic Properties of GMM estimator

Consistency of GMM estimator

Substitution of y by $y = Z\theta_0 + u$ in the GMM estimator gives

$$\hat{\theta} = (Z' X W X' Z)^{-1} Z' X W X' y = \theta_0 + Q \bar{m}, \quad Q \equiv T(Z' X W X' Z)^{-1} Z' X W$$

Assume that $W_0 = \text{plim}(W)$ and $\Sigma_{xx} = \text{plim}(T^{-1} Z' X) = \Sigma_{xx}'$ are finite matrices with a full row rank. Then,

$$\bar{Q} \equiv \text{plim}(Q) = (\Sigma_{xx} W_0 \Sigma_{xx})^{-1} \Sigma_{xx} W_0$$

and hence,

$$\text{plim}(\hat{\theta}) = \theta_0 + \bar{Q} \text{plim}(\bar{m})$$

Recall that

$$\bar{m} = T^{-1} X' u = \frac{1}{T} \sum_{t=1}^T X_t' u_t = \frac{1}{T} \sum_{t=1}^T m_t$$

If m_t is independent, the standard Law of Large Numbers (LLN) applies so that $\text{plim}(\bar{m}) = E(X_t' u_t) = 0$. This gives $\text{plim}(\hat{\theta}) = \theta_0$. If m_t is not independent, the standard LLN does not apply, and we need a generalization of the LLN to a sequence of dependent random variables. One such theorem is the *ergodic theorem* which requires that the autocorrelation between m_t and m_{t-s} goes to zero as s goes to infinity.

Asymptotic Distribution of GMM estimator

Note that

$$\sqrt{T}(\hat{\theta} - \theta_0) = Q \sqrt{T} \bar{m}(\theta_0)$$

Therefore, the asymptotic distribution of $\hat{\theta}$ depends on the asymptotic distribution of $\sqrt{T} \bar{m}(\theta_0)$. If m_t is a sequence of independent identically distributed random variables, the standard Lindberg-Levy central limit theorem applies. However, if they are independent, but not identically distributed, we need the Lindberg-Feller central limit theorem, which is a generalization of the Lindberg-Levy CLT. If they are not independent nor identically distributed, we need a more general central limit theorem such as the ergodic central limit theorem. Each case requires a different set of assumptions about the stochastic process of m_t . But, they all boil down to the conditions such that the following convergence in distribution holds:

$$\sqrt{T}(\bar{m}(\theta_0) - \bar{\mu}(\theta_0)) = \sqrt{T} \bar{m}(\theta_0) \xrightarrow{d} N(0, S_0)$$

where

$$S_0 = \text{plim} \left(\frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T X_t' u_t u_s' X_s \right) = \text{plim} \left(\frac{1}{T} X' \Omega X \right)$$

If this holds, then

$$\sqrt{T}(\hat{\theta} - \theta_0) = Q\sqrt{T}\bar{m}(\theta_0) \xrightarrow{d} N(0, \bar{Q}S_0\bar{Q}')$$

Theorem. Suppose that $W_0 = \text{plim}(W)$ and $\Sigma_{zx} = \text{plim}(T^{-1}Z'X) = \Sigma_{zx}'$ are finite matrices with a full row rank. Then, under some additional assumptions about the stochastic properties of m_t , the GMM estimator $\hat{\theta}$ is consistent and asymptotically normally distributed:

$$\text{plim}(\hat{\theta}) = \theta$$

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$$

where

$$V = \bar{Q}S_0\bar{Q}' = (\Sigma_{zx}W_0\Sigma_{zx})^{-1}\Sigma_{zx}W_0S_0W_0\Sigma_{zx}(\Sigma_{zx}W_0\Sigma_{zx})^{-1}$$

$$S_0 = \text{plim}\left(\frac{1}{T}X'\Omega X\right)$$

Efficient GMM Estimator

To implement the GMM, we need to choose the weighting matrix W . What is the optimal W ? The optimal weighting matrix is the one that leads to the smallest asymptotic variance of GMM estimator of θ .

Lemma. Let V_e be the covariance matrix of the GMM estimator of θ when $W_0 = S_0^{-1}$. Then,

$$V - V_e = (\Sigma_{zx}W_0\Sigma_{zx})^{-1}\Sigma_{zx}W_0S_0W_0\Sigma_{zx}(\Sigma_{zx}W_0\Sigma_{zx})^{-1} - (\Sigma_{zx}W_0\Sigma_{zx})^{-1} \succeq 0$$

is positive semidefinite for any other $W_0 \neq S_0^{-1}$.

Homework #4

(c) Prove the *Lemma* stated above.

This Lemma suggests that the optimal choice of W is the one which converges to S_0^{-1} in probability. Since $S_0 = \text{plim}(X'\Omega X/T)$, the optimal weighting matrix is given by

$$W_e = (X'\Omega X/T)^{-1}$$

The objective function of the GMM with this choice of W becomes

$$J_e(\theta) = T[\bar{m}(\theta)]' W_e [\bar{m}(\theta)] = T[T^{-1} X' (y - Z\theta)]' (X' \Omega X / T)^{-1} [T^{-1} X' (y - Z\theta)]$$

$$= (y - Z\theta)' X (X' \Omega X)^{-1} X' (y - Z\theta)$$

The first order condition with respect to θ is

$$Z' X [X' \Omega X]^{-1} X' (y - Z\theta) = 0$$

and the solution for θ for a given Ω is

$$\hat{\theta}_e = [Z' X (X' \Omega X)^{-1} X' Z]^{-1} Z' X (X' \Omega X)^{-1} X' y$$

This is the efficient GMM estimator and has an asymptotic normal distribution

$$\sqrt{T}(\hat{\theta}_e - \theta) \xrightarrow{d} N(0, V_e), \quad V_e = (\Sigma_{xx} W_0 \Sigma_{xz})^{-1}$$

V_e is estimated by

$$\hat{V}_e = (\hat{\Sigma}_{xx} \hat{W}_0 \hat{\Sigma}_{xz})^{-1} = \left(\frac{Z' X}{T} \left(\frac{X' \Omega X}{T} \right)^{-1} \frac{X' Z}{T} \right)^{-1} = T [Z' X (X' \Omega X)^{-1} X' Z]^{-1}$$

and the asymptotic covariance matrix of $\hat{\theta}_e$ is computed by $cov(\hat{\theta}_e) = [Z' X (X' \Omega X)^{-1} X' Z]^{-1}$.

Remarks:

Premultiply X' to the linear regression model under study to get

$$X' y = X' Z \theta + \epsilon, \quad \epsilon = X' u, \quad cov(\epsilon) = X' \Omega X$$

The efficient GMM estimator is the GLS estimator of this transformed regression model. Hence, we can also compute the efficient GMM estimator of θ by the OLS from

$$P X' y = P X' Z \theta + P \epsilon \Rightarrow y^* = Z^* \theta + \epsilon^*$$

where $P' P = (X' \Omega X)^{-1}$, $y^* = P X' y$, $Z^* = P X' Z$ and $\epsilon^* = P X' u$:

$$\hat{\theta} = (Z^{*'} Z^*)^{-1} Z^{*'} y^*, \quad cov(\hat{\theta}) = (Z^{*'} Z^*)^{-1}$$

Feasible Efficient GMM Estimator

The expression for the estimator $\hat{\theta}_e$ is written as if Ω is known. If Ω is unknown, we need to estimate it along with θ such that the first order condition

$$Z' X [X' \hat{\Omega} X]^{-1} X' (y - Z\hat{\theta}) = 0$$

is satisfied. We will discuss two methods: (A) estimate Ω and θ in two steps (or repetition of two steps), and (B) estimate Ω and θ simultaneously.

(A) Two-Step Estimator

Step 1. Estimate Ω .

Compute the initial consistent GMM estimator $\tilde{\theta}$ by setting $W=(X'X/T)^{-1}$. This is equivalent to assuming $\Omega=I$, and the resulting estimator is the standard 2SLS estimator.

Compute $\tilde{u}=y-Z\tilde{\theta}$, and then compute $\hat{\Omega}$ from $\tilde{u}$ by using an appropriate method.

Step 2. Estimate the efficient GMM estimator by using $\hat{\Omega}$

The estimator $\hat{\Omega}$ in the first step can be the heteroscedasticity consistent (HC) estimator $\hat{\Omega}_{HC}(\tilde{\theta}) = \text{diag}(\tilde{u}_t^2)$, or one of the heteroscedasticity-autocorrelation consistent (HAC) estimators

$$\text{Hansen-White: } \hat{S}_{hw} = \sum_{j=-q}^q \hat{\Gamma}_j, \quad \hat{\Gamma}_j = \frac{1}{T} \sum_{t=j+1}^T X_t' \tilde{u}_t \tilde{u}_{t-j} X_{t-j},$$

$$\text{Newey-West: } \hat{S}_{nw} = \hat{\Gamma}_0 + \sum_{j=1}^q \left(1 - \frac{j}{1+q}\right) (\hat{\Gamma}_j + \hat{\Gamma}_j')$$

Repeated Two-Step Estimator

One may repeat the two-step estimator by estimating Ω based on the residuals from second step estimator $\hat{u}_e = y - Z\hat{\theta}_e$. Repetition may continue until the changes of the estimator of θ is within a reasonable tolerance level. However, the asymptotic properties of the estimators remain the same regardless of the iteration, though finite sample estimates will be different.

(B) Continuous Updating Estimator

Ω is estimated from the estimates of error terms, and error terms are estimated from the estimator of θ . Therefore, an estimator of Ω is an estimator of $\Omega(\theta)$. For example, the HC estimator is an estimator of

$$\Omega(\theta) = \text{diag}(u_t^2) = \text{diag}((y_t - Z_t\theta)^2)$$

The two step estimator evaluates this at the initial estimator $\tilde{\theta}$ and then computes the estimate of θ to satisfy the first order condition

$$Z'X[X'X\hat{\Omega}(\tilde{\theta})X]^{-1}X'(y - Z\hat{\theta}_e) = 0$$

This shows that the estimator of θ that enters in the estimator of Ω and the final estimator of θ are different.

The continuous updating estimator which has been proposed more recently finds an estimator $\hat{\theta}_c$ such that

$$Z'X[X'X\hat{\Omega}(\hat{\theta}_c)X]^{-1}X'(y - Z\hat{\theta}_c) = 0$$

The continuous updating estimator $\hat{\theta}_c$ is thus a solution of a highly nonlinear first order condition.

Statistical Inference in GMM estimation

A hypothesis about the coefficient parameters can be tested by using the same type of intuition as the likelihood ratio, Wald and Lagrange Multiplier (score) tests. A hypothesis that is important in the GMM estimation is the question related to the validity of the moment conditions, which are used as the building block. We will discuss this first below.

Test of Overidentifying Moment Restrictions

The GMM estimator is based on the population specified moment conditions. Are they all valid? Is it possible that some of them are invalid? Can we test their validity?

We cannot test the validity of all moments because we need at least minimum number of just identifying moment condition for identifiability of parameters. When there are more moment conditions than needed to determine parameters, we can test whether extra moment conditions are valid. The null and alternative hypotheses are

$$H_0: \mu_t(\theta_0) = E(X_t' u_t) = 0 \text{ for all } t, \quad H_1: H_0 \text{ is false}$$

The null hypothesis can also be expressed as

$$H_0: \bar{\mu}(\theta_0) = 0, \quad \bar{\mu}(\theta_0) = T^{-1} \sum_{t=1}^T \mu_t(\theta_0)$$

The null hypothesis says that all moment conditions are valid. The alternative hypothesis says that at least one of the extra moment conditions is false, though it does not specify which ones are taken as just-identifying conditions. When the null hypothesis is rejected, we do not know which moment condition caused the rejection. We simply conclude that the data does not support the statement that all moment conditions are correct.

Let $\hat{\theta}$ be the efficient GMM estimator with the optimum weighting matrix $W_e = (X' \Omega X / T)^{-1}$. The test statistic for the test of overidentifying moment conditions is

$$J(\hat{\theta}) = T[\bar{m}(\hat{\theta})]' W_e [\bar{m}(\hat{\theta})] \xrightarrow{d} \chi^2(K-p) \text{ under } H_0$$

This is the criterion (objective) function of the efficient GMM, and it is called **Hansen's J-statistic** of overidentifying restrictions (Hansen, 1982). The null hypothesis is rejected if $J(\hat{\theta})$ is greater than the critical value $\chi^2_{\alpha}(K-p)$ of significance level α .

The asymptotic null distribution is easily seen from

$$\sqrt{T}(\bar{m}(\theta_0) - \bar{\mu}(\theta_0)) \xrightarrow{d} N(0, S_0)$$

and the distribution of its quadratic form

$$T[\bar{m}(\theta_0) - \bar{\mu}(\theta_0)]' S_0^{-1} [\bar{m}(\theta_0) - \bar{\mu}(\theta_0)] \xrightarrow{d} \chi^2(K-p)$$

which becomes $T[\bar{m}(\theta_0)]' S_0^{-1} [\bar{m}(\theta_0)]$ under the null hypothesis $\bar{\mu}(\theta_0)=0$. Since the efficient GMM estimator is a consistent estimator and $plim(W_e)=S_0^{-1}$, Hansen's J -statistic converges to the same $\chi^2(K-p)$.

Notice that the J -statistic is a *Wald type* test statistic as it is based on the distribution of unrestricted estimator $\bar{m}(\hat{\theta})$ of $\bar{\mu}(\theta_0)$. It can also be interpreted as a Likelihood Ratio type test statistic which compares the values of the objective function evaluated at restricted and unrestricted estimators. Let J_K be the minimum value of the GMM objective function under all K moment conditions. This is the restricted value of the objective function, where the restrictions refer to the extra overidentifying restrictions beside the just identified restrictions which are required for parameter identifiability. Let J_p be the minimum value of the objective function under p just-identifying restrictions, which are a subset of all moment conditions. It does not matter which subset we choose. What is the value of J_p ? Since it is a system of p moment equations in p parameters, a perfect fit is obtained and $J_p=0$. And $J_K - J_p = J_K$ which has a $\chi^2(K-p)$ distribution.

The reason for the loss of degree of freedom is that, by the first order conditions of the GMM estimator, p linear combinations of the sample moments are zeros for every sample. Therefore, there are only $K-p$ non-degenerate random variables in $\bar{m}(\hat{\theta})$. This can be shown more explicitly in a linear regression model. The minimum value of the objective function evaluated at the efficient GMM estimator with $W_e=(X'\Omega X/T)^{-1}$ can be written as $J(\hat{\theta}) = (C^{-1}u)'B(C^{-1}u)$, where $\Omega=CC'$ and B is a symmetric idempotent matrix with rank $K-p$. Since $C^{-1}u$ is a standard normal vector, $J(\hat{\theta})$ converges to $\chi^2(K-p)$.

Homework #4

(d) Show that the minimum value of the objective function evaluated at the efficient GMM estimator with $W_e=(X'\Omega X/T)^{-1}$ can be written as $J(\hat{\theta}) = (C^{-1}u)'B(C^{-1}u)$, where $\Omega=CC'$. Show that B is a symmetric idempotent matrix with rank $K-p$.

Test of a Subset of Overidentifying Moment Restrictions

The full set test of overidentifying restrictions tests the null hypothesis $H_0: E(X_t' u_t) = 0$. The subset test considers the validity of orthogonality of a prechosen subset of X_t , while the rest of the moment restrictions are maintained to be true. Partition the moment conditions as

$$E \begin{pmatrix} X_{t1}' u_t \\ X_{t2}' u_t \end{pmatrix} = \begin{pmatrix} \mu_1(\theta_0) \\ \mu_2(\theta_0) \end{pmatrix}$$

where X_{ti}' and μ_i are $1 \times K_i$ vectors. The full set test the hypothesis $\mu_1(\theta_0)=0$ and $\mu_2(\theta_0)=0$. For the subset test, we maintain $\mu_2(\theta_0)=0$ under the assumption that $K_2 \geq p$ so that the moment conditions are just- or overidentified under the maintained hypothesis. We then test $H_0: \mu_1(\theta)=0$. We may use the Wald type test based on the asymptotic distribution of $\bar{m}_1(\hat{\theta})$

$$J_s(\hat{\theta}) = T[\bar{m}_1(\hat{\theta})]' W_{e,11} [\bar{m}_1(\hat{\theta})] \xrightarrow{d} \chi^2(K_1) \text{ under } H_0$$

where $W_{e,11}$ is the $K_1 \times K_1$ submatrix of W_e corresponding to $\bar{m}_1$.

We may also use the LR type test. Let $J(\hat{\theta})$ be the estimated value of the objective function of efficient GMM with all moment conditions (i.e., with $\mu_1(\theta) = \mu_2(\theta) = 0$), and $J_2(\tilde{\theta})$ be the estimated value of the objective function of efficient GMM under only $\mu_2(\theta) = 0$. Then the difference $J(\hat{\theta}) - J_2(\tilde{\theta})$ has a chi-square distribution

$$J(\hat{\theta}) - J_2(\tilde{\theta}) \xrightarrow{d} \chi^2(K_1) \text{ under } H_0$$

If this statistic exceeds the critical value $\chi^2_{\alpha}(K_1)$, then we reject the null hypothesis.

Remark: An efficient GMM estimator requires an estimation of weighting matrix W . The estimator of W will be different for estimators $\hat{\theta}$ and $\tilde{\theta}$. This may yield a non-positive test statistic in a finite sample. Therefore, the weight matrix for the estimator $\tilde{\theta}$ is often chosen as a submatrix $\hat{W}_{11}$ of $\hat{W}$, which is the matrix for all moment conditions.

Test of Linear Restrictions on Coefficient Parameters

Consider the test of a linear hypothesis $H_0: R\theta = q$ against alternative hypothesis $H_1: R\theta \neq q$, where R is a $r \times p$ matrix of known constants with a full row rank, $\text{rank}(R) = r$, and q is a $r \times 1$ vector of known constants. This test can be conducted by using the Wald type, LR type and LM type tests. The latter two tests require estimators under the restrictions of the null hypothesis.

The two-step GMM estimator requires a consistent estimator of Ω , and it can be estimated from the residuals of the unrestricted and restricted initial estimator of θ . Let $\hat{\Omega}$ and $\tilde{\Omega}$ be the unrestricted and restricted estimator of Ω , respectively. For an easy reference, a few notations are collected below:

$$A = Z'X(X'\Omega X)^{-1}X'Z, \quad b = Z'X[X'\Omega X]^{-1}X'y, \quad C = RA^{-1}R'$$

$$V = (\Sigma_{xx}W_0\Sigma_{xx})^{-1} = \text{plim}(TA^{-1}), \quad W_0 = \text{plim}(W) = \text{plim}(X'\Omega X/T)^{-1}$$

We will use TA^{-1} as the estimator of V . The estimator of A is denoted by $\hat{A}$ if $\hat{\Omega}$ is used and by $\tilde{A}$ if $\tilde{\Omega}$ is used. And similar notations are used for the estimators of b , C , V and W .

The unrestricted GMM estimator is given by

$$\hat{\theta} = A^{-1}b$$

where A and b are estimated with $\hat{\Omega}$, and it has an asymptotic normal distribution

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$$

and the asymptotic covariance matrix is estimated by $\hat{V} = T\hat{A}\hat{A}^{-1}$. It is straightforward to derive the restricted estimators of θ and the Lagrange multiplier λ :

$$\tilde{\theta} = A^{-1}b - A^{-1}R'C^{-1}RA^{-1}b + A^{-1}R'C^{-1}q = A^{-1}b - A^{-1}R'C^{-1}(RA^{-1}b - q)$$

$$\tilde{\lambda} = C^{-1}RA^{-1}b - C^{-1}q$$

where matrices A , b and C are estimated by restricted $\tilde{\Omega}$. It is straightforward to show

$$\frac{1}{\sqrt{T}} \tilde{\lambda} \xrightarrow{d} N(R\theta - q, (RVR')^{-1})$$

Wald type test

The Wald type test examines whether the unrestricted estimator satisfies the restrictions under the null hypothesis (i.e., $R\hat{\theta} \approx q$). If the restrictions are true, then the unrestricted estimator $R\hat{\theta}$ will be close to q . The asymptotic normal distribution of the unrestricted efficient GMM estimator

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, V)$$

implies

$$\sqrt{T}(R\hat{\theta} - q) \xrightarrow{d} N(R\theta - q, RVR')$$

Hence, its quadratic form has an asymptotic χ^2 distribution

$$WT = T(R\hat{\theta} - q)'(RVR')^{-1}(R\hat{\theta} - q) \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

We use $\hat{V}$ as the estimator of V so that the WT test statistic becomes

$$WT = (R\hat{\theta} - q)'(\hat{A}\hat{A}'^{-1}R')^{-1}(R\hat{\theta} - q)$$

and reject the null hypothesis if WT is greater than the critical value.

Likelihood Ratio type test

This test compares the values of the objective function evaluated at the constrained and unconstrained estimators. If the restrictions are true, restricted and unrestricted values of the objective function will roughly equal. Since the objective function of the efficient GMM estimator is

$$J(\theta) = (y - Z\theta)'X(X'\Omega X)^{-1}X'(y - Z\theta)$$

Let $J_r(\tilde{\theta})$ and $J_u(\hat{\theta})$ be the minimum values of the objective function at restricted and unrestricted estimators, respectively. That is,

$$J_r(\tilde{\theta}) = (y - Z\tilde{\theta})'X(X'\tilde{\Omega}X)^{-1}X'(y - Z\tilde{\theta})$$

$$J_u(\hat{\theta}) = (y - Z\hat{\theta})'X(X'\hat{\Omega}X)^{-1}X'(y - Z\hat{\theta})$$

Then,

$$LRT = J_r(\tilde{\theta}) - J_u(\hat{\theta}) \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

This test statistic may not be positive because of the different estimator of Ω . Since what we need is a consistent estimator of Ω , it is a common practice to use the unrestricted estimator $\hat{\Omega}$ in the restricted estimation of θ also partly because it is consistent under both the null and alternative hypotheses and partly because it is cumbersome to compute two different estimator of Ω . Let the restricted estimator of θ by using $\hat{\Omega}$ be denoted by

$$\tilde{\theta}_u = \hat{A}^{-1}\hat{b} - \hat{A}^{-1}R'\hat{C}^{-1}R\hat{A}^{-1}\hat{b} + \hat{A}^{-1}R'\hat{C}^{-1}q = \hat{\theta} - \hat{A}^{-1}R'\hat{C}^{-1}(R\hat{\theta} - q)$$

Then, the objective function becomes

$$J_r(\tilde{\theta}_u) = (y - Z\tilde{\theta}_u)'X(X'\hat{\Omega}X)^{-1}X'(y - Z\tilde{\theta}_u)$$

and the LR type test statistic is

$$LRT_u = J_r(\tilde{\theta}_u) - J_u(\hat{\theta}) \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

This statistic is identical to the WT test statistic.

Lagrange Multiplier type test

The LM type test examines whether the estimator of Lagrange multiplier is close to zero or not. If the null hypothesis is true, $\tilde{\lambda}$ will be close to zero. Since $\tilde{\lambda}$ has an asymptotic normal distribution, its quadratic form has a χ^2 distribution

$$LMT = \frac{1}{T} \tilde{\lambda}'(RVR')\tilde{\lambda} \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

If V is estimated by the restricted estimator $\tilde{V}(\tilde{\Omega}) = T\tilde{A}^{-1}$, then LMT becomes

$$LMT = \tilde{\lambda}'(R\tilde{A}^{-1}R')\tilde{\lambda} \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

Substituting $R'\tilde{\lambda} = -(\tilde{A}\tilde{\theta} - \tilde{b})$ from the first order condition of the restricted minimization, this statistic can be written in the form of the score test

$$LMT = (\tilde{A}\tilde{\theta} - \tilde{b})'\tilde{A}^{-1}(\tilde{A}\tilde{\theta} - \tilde{b}) \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

If the unrestricted estimator $\hat{\Omega}$ is used for the second-step restricted estimation, this test statistic becomes

$$LMT_u = (\hat{A}\hat{\theta}_u - \hat{b})'\hat{A}^{-1}(\hat{A}\hat{\theta}_u - \hat{b}) \xrightarrow{d} \chi^2(r) \text{ under } H_0$$

Homework #4

(e) Derive $\tilde{\theta}$ and $\tilde{\lambda}$ from

$$\min J(\theta) = (y - Z\theta)'X(X'\Omega X)^{-1}X'(y - Z\theta) \quad \text{subject to } R\theta = q$$

(f) Show that WT , LRT_u and LMT_u are identical.

Appendix
Generalized Method of Moments Estimator
nonlinear moment conditions

Suppose that there are K population moment conditions. The population moment conditions and the corresponding sample moment conditions are denoted by

$$\mu_i(\theta_0) \equiv E[m(z_p, \theta_0)] = E \begin{pmatrix} m_1(z_p, \theta_0) \\ m_2(z_p, \theta_0) \\ \vdots \\ m_K(z_p, \theta_0) \end{pmatrix} = 0 \quad \dots \text{population moment conditions}$$

$$\bar{m}(\theta) = \frac{1}{T} \sum_{t=1}^T m(z_p, \theta) = 0 \quad \dots \text{sample moment conditions}$$

where z_t is a vector of stationary ergodic random variables and θ_0 is a p dimensional true parameters with $p \leq K$. For the parameters to be *identifiable*, there must be a unique θ_0 at which the population moment conditions are satisfied.

When the moment conditions are overidentified ($p < K$), the generalized method of moment (GMM) estimator minimizes the criterion function (*distance function*)

$$\min_{\theta} J(\theta) = T[\bar{m}(\theta)]' W[\bar{m}(\theta)]$$

where W is a $K \times K$ symmetric positive definite weighting matrix which does *not* depend on θ . Let $D(\theta)$ be the $K \times p$ matrix of derivatives

$$D(\theta) = \frac{\partial \bar{m}(\theta)}{\partial \theta'} = \left(\frac{\partial \bar{m}_i(\theta)}{\partial \theta_j} \right) = \begin{pmatrix} \frac{\partial \bar{m}_1(\theta)}{\partial \theta_1} & \dots & \frac{\partial \bar{m}_1(\theta)}{\partial \theta_p} \\ \vdots & & \vdots \\ \frac{\partial \bar{m}_K(\theta)}{\partial \theta_1} & \dots & \frac{\partial \bar{m}_K(\theta)}{\partial \theta_p} \end{pmatrix}$$

The first order conditions are

$$\frac{\partial J}{\partial \theta} = 2TD'(\theta)W\bar{m}(\theta) = 0 \Rightarrow D'(\hat{\theta})W\bar{m}(\hat{\theta}) = 0$$

The last expression shows that the GMM estimator uses only p linear combinations of sample moment conditions where the selection matrix is a $p \times K$ matrix $D'(\hat{\theta})W$. The first order conditions can be highly nonlinear in θ and may need to be solved numerically. When the moment conditions are linear in unknown

parameters, the closed form solution exists.

Asymptotic Properties of GMM estimator

Under a set of relatively mild conditions (including conditions $\text{plim} W = W_0$ and $\text{plim} \bar{m}(\theta) = \mu(\theta)$), the GMM estimator $\hat{\theta}$ is a consistent estimator, $\text{plim}(\hat{\theta}) = \theta_0$, and it converges to a normal distribution

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$$

To show the asymptotic distribution, take the first order Taylor expansion of the sample moment condition around the true value θ_0 :

$$\bar{m}(\hat{\theta}) = \bar{m}(\theta_0) + \frac{\partial \bar{m}(\theta_1)}{\partial \theta'} (\hat{\theta} - \theta_0) = \bar{m}(\theta_0) + D(\theta_1)(\hat{\theta} - \theta_0)$$

where θ_1 is between $\hat{\theta}$ and θ_0 . Premultiplication of $D'(\hat{\theta})W$ gives

$$D'(\hat{\theta})W\bar{m}(\hat{\theta}) = 0 = D'(\hat{\theta})W\bar{m}(\theta_0) + D'(\hat{\theta})WD(\theta_1)(\hat{\theta} - \theta_0)$$

where the first equality is due to the first order conditions. Rearranging the terms, we can write

$$\sqrt{T}(\hat{\theta} - \theta_0) = -[D'(\hat{\theta})WD(\theta_1)]^{-1}D'(\hat{\theta})W\sqrt{T}\bar{m}(\theta_0) \equiv Q\sqrt{T}\bar{m}(\theta_0)$$

Since $\bar{m}(\theta_0)$ is the sample mean, the central limit theorem gives

$$\sqrt{T}(\bar{m}(\theta_0) - \mu(\theta_0)) = \sqrt{T}\bar{m}(\theta_0) \xrightarrow{d} N(0, S_0)$$

where $\bar{\mu}(\theta_0) = T^{-1} \sum_{i=1}^T \mu_i(\theta_0)$. Therefore,

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V), \quad V = [\text{plim} Q] S_0 [\text{plim} Q]'$$

where

$$\begin{aligned} \text{plim} Q &= -[\text{plim} D'(\hat{\theta}) \times \text{plim}(W) \times \text{plim} D(\theta_1)]^{-1} \text{plim} D'(\hat{\theta}) \times \text{plim} W \\ &= -(D_0' W_0 D_0)^{-1} D_0' W_0 \end{aligned}$$

which gives

$$V = (D_0' W_0 D_0)^{-1} (D_0' W_0 S_0 W_0 D_0) (D_0' W_0 D_0)^{-1}$$